Yu Li

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Education

George Washington University — **Washington, D.C.**
Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0 — *2025.08 – 2029.05 (expected)*
Advisor: Prof. Tian Lan; Work with: Prof. Zhengling Qi

Wuhan University, Hongyi Honor College — **Wuhan, China**
B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0 — *2021.09 – 2025.06*

Experience

ByteDance, Tiktok — **San Jose, CA**
Research Scientist Intern, Recommendation Core — *2026.05 – 2026.08*

- Owned the ranking stage of TikTok video recommendation, built on a 3B-parameter Large Recommendation Model.
- Adapted LLM architecture techniques to the ranking model, including multi-head latent attention, time-aware RoPE, and mixed-query optimization.
- Carried the model from offline training to production deployment, with gains on core business metrics.

Research Highlights

LLM Reasoning	Post-Training, Inference Efficiency	[NeurIPS'26a] [NeurIPS'26b] [COLM'26] [P2] [P3]
Agentic Systems	Long-Horizon Agent, Multi-Agent Systems	[ACL'26] [INFOCOM'26] [P1]
Vision and Control	Generative Models, Decision-Making	[CVPR'26] [INFOCOM'26] [P4]

Preprints & Under Review

P=Preprint

[P1] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning**
Yu Li, Rui Miao, Zhengling Qi, Tian Lan

[P2] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO**
Yu Li, Tian Lan, Zhengling Qi

[P3] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning**
Yu Li, Shu Hong, Tian Lan
Under review. ARR May 2026 OA: 3.33/5

[P4] **VideoGen-Agent: Reinforcing Video Generation Agents**
Binxu Li, Haoyi Duan, Yuhui Zhang, Yaohui Zhang, Zihao Lin, Kaituo Feng, Suozhi Huang, Xiangyi Li, **Yu Li**, Chunyuan Li, Shilong Liu, Mengdi Wang

Selected Publications [Full List]

†=Equal Contribution

[NeurIPS'26a] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning**
Yu Li, Rui Miao, Tian Lan, Zhengling Qi

[NeurIPS'26b] INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization
Yu Li, Tian Lan, Zhengling Qi
ICLR 2026 Workshop on AI with Recursive Self-Improvement

[COLM'26] MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference
Yu Li, Binxu Li, Tian Lan

[ACL'26 Findings] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization
Yu Li, Sizhe Tang, Tian Lan

[CVPR'26] CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion
Yu Li, Yujun Cai, Chi Zhang

[ICASSP'26 Oral] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models
Yu Li, Da Chang, Xi Xiao

[ACM MM'26] Rethinking Layer-Wise Information Allocation for Vision Foundation Model Adaptation
Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, **Yu Li**, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian

[AAAI'26] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT
Da Chang, Peng Xue, **Yu Li**, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

[INFOCOM'26] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense
Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

Selected Awards & Honors

- **ICML 2026 Gold Reviewer**, top reviewers recognition. *2026*
- **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. *2024*
- **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. *2023, 2024*
- **First-Class Scholarship**, Top 5%, ¥3,000, three times. *2022, 2023, 2024*

Academic Services

- **Conference Reviewer:** *2027:* AAAI. *2026:* NeurIPS, ICML, ICLR, COLM, AAAI, ACM MM.
- **Journal Reviewer:** TPAMI, TMLR, ToN, Neurocomputing.

Skills

- **Training Frameworks:** PyTorch, verl, Hugging Face Transformers / Accelerate / TRL, Hydra.
- **Inference & Distributed:** vLLM, Ray, FSDP, NCCL multi-node training, Docker, Linux GPU clusters (SSH, tmux).
- **Languages:** English (Proficient), Mandarin (Native), Japanese (Basic).